

Geophysical Characterization of the Historic Onyx Mine in Steamboat Springs, CO

Non-invasive geophysical techniques can be used to classify and/or identify subsurface structures like mine shafts. This is especially important in locations where the collapse of such mine shafts would pose a risk to public safety. Changes in mass distribution, like a mine shaft, affect how signals propagate through the subsurface, and the methods used in this survey are sensitive to these variations. Ground penetrating radar (GPR) is sensitive to dielectric permittivity, conductivity, and magnetic permeability of the material through which its signals propagate. Gravimeters measure the acceleration due to gravity and are sensitive to density variations. A void in the subsurface would result in a negative gravitational acceleration anomaly, and GPR would show us shallow-subsurface reflectors. A single geophysical method's result is non-unique; ideal survey designs use multiple methods in tandem to increase the confidence of a survey's result. The gravity results show a negative residual anomaly where we expect to see an established mine shaft. It became apparent that the terrain, instead of the void, was largely influencing our gravity anomalies after applying advanced terrain corrections. GPR did not provide any information about the mine shaft; our GPR instruments had relevant data down to 100 ns, while mine shaft reflectors would be seen between 80-200 ns. We cannot claim unique results from these geophysical techniques alone, but we can still make interpretations on the mine shaft with the addition of outside resources such as literature. The use of gravimeters and GPR aimed to classify a known mine shaft; however, due to limitations in survey designs and a lithology that was not conducive to GPR, we cannot claim uniqueness of our results. With more intensive survey parameters, denser gravimeter coverage, and the addition of electrical resistivity tomography (ERT), the mine shaft structure could be identified and ultimately characterized for public safety uses.